examples, various outcomes—heads versus tails, red versus black, one lottery number versus another—are assigned probabilities with the understanding that one or another result will definitely happen and that each result is the end product of an independent, definite history. When a coin is tossed, sometimes the spinning motion is just right for the toss to come out heads and sometimes it's just right for the toss to come out tails. The 50-50 probability we assign to each outcome refers not just to the final result—heads or tails— but also to the histories that lead to each outcome. Half of the possible ways you can toss a coin result in heads, and half result in tails. The histories themselves, though, are totally separate, isolated alternatives. There is no sense in which different motions of the coin reinforce each other or cancel each other out. They're all independent.

But in quantum mechanics, things are different. The alternate paths an electron can follow from the two slits to the detector are not separate, isolated histories. The possible histories commingle to produce the observed outcome. Some paths reinforce each other, while others cancel each other out. Such quantum interference between the various possible histories is responsible for the pattern of light and dark bands on the detector screen. Thus, *the telltale difference between the quantum and the classical notions of probability is that the former is subject to interference and the latter is not.*

Decoherence is a widespread phenomenon that forms a bridge between the quantum physics of the small and the classical physics of the not-so-small by suppressing quantum interference—that is, by diminishing sharply the core difference between quantum and classical probabilities. The importance of decoherence was realized way back in the early days of quantum theory, but its modern incarnation dates from a seminal paper by the German physicist Dieter Zeh in 1970,[14] and has since been developed by many researchers, including Erich Joos, also from Germany, and Wojciech Zurek, of the Los Alamos National Laboratory in New Mexico.

Here's the idea. When Schrödinger's equation is applied in a simple situation such as single, isolated photons passing through a screen with two slits, it gives rise to the famous interference pattern. But there are two very special features of this laboratory example that are not